

MAJOR PROJECT DOCUMENTATION

SECOND BRAIN

A Fully Local, Fine-Tuned Personal RAG System over Browsing History

Chrome Extension (Manifest V3) • Local RAG Pipeline • Fine-Tuned Local LLM (SFT + DPO)

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1. Executive Summary

Second Brain is a privacy-first, fully local personal knowledge retrieval system built as a Chrome extension paired with a local Retrieval-Augmented Generation (RAG) pipeline and a purpose-built, fine-tuned language model. It passively indexes the web pages a person actually reads, builds a searchable, cited knowledge base from that reading history, and answers natural-language questions using only content the user has genuinely encountered — explicitly refusing to answer, rather than guessing, when the requested information is not present in the user's own history.

The project is deliberately scoped as an industry-grade system rather than a course assignment. It runs entirely on the user's device with zero cloud dependency and zero recurring cost, and it is validated against a real, uncurated two-week corpus of the developer's own browsing — the messiest and most realistic dataset available, since no public benchmark can anticipate what a specific person will read in a given fortnight. The core technical contribution beyond the base retrieval pipeline is a small open-weight language model fine-tuned specifically for grounded, citation-backed answering and calibrated refusal, trained using Supervised Fine-Tuning (SFT) followed by Direct Preference Optimization (DPO) on preference pairs mined directly from the system's own documented failures.

2. Problem Statement

Knowledge workers, researchers, and students routinely read far more than they can recall. A meaningful fraction of what is read — including material that later becomes relevant — is functionally lost within days, buried in an unstructured, non-semantic browser history that records only URLs and timestamps, not meaning.

Existing solutions fall into two unsatisfying categories:

- Manual tools (bookmarks, read-later apps, notes) require active curation discipline that most people do not maintain consistently.
- Cloud-based “AI memory” browser tools solve retrieval and generation, but introduce ongoing infrastructure cost, real privacy exposure (a third party gains a durable record of everything a person reads), and rely on general-purpose language models that are not optimized for this task's specific failure mode.

That specific failure mode is the core problem this project addresses. General-purpose LLMs are typically optimized to be maximally helpful and confident, which makes them prone to producing plausible-sounding but fabricated answers when the true answer is unknown or absent from context. For a system positioned as a faithful personal memory, this is the worst possible failure: a confident wrong answer is more damaging than an honest “not found.”

- **Problem statement:** Can a fully local, zero-cost system faithfully reconstruct and retrieve what a specific person actually read, and reliably distinguish what it knows from what it does not, using a generation model fine-tuned specifically for grounded answering and calibrated refusal over a real, messy, personal corpus?

3. Proposed Solution

Second Brain answers this with two cooperating layers that together form a closed loop: capture → clean → dedupe → index → retrieve → generate → cite, running entirely on-device.

3.1 Capture & Retrieval Layer

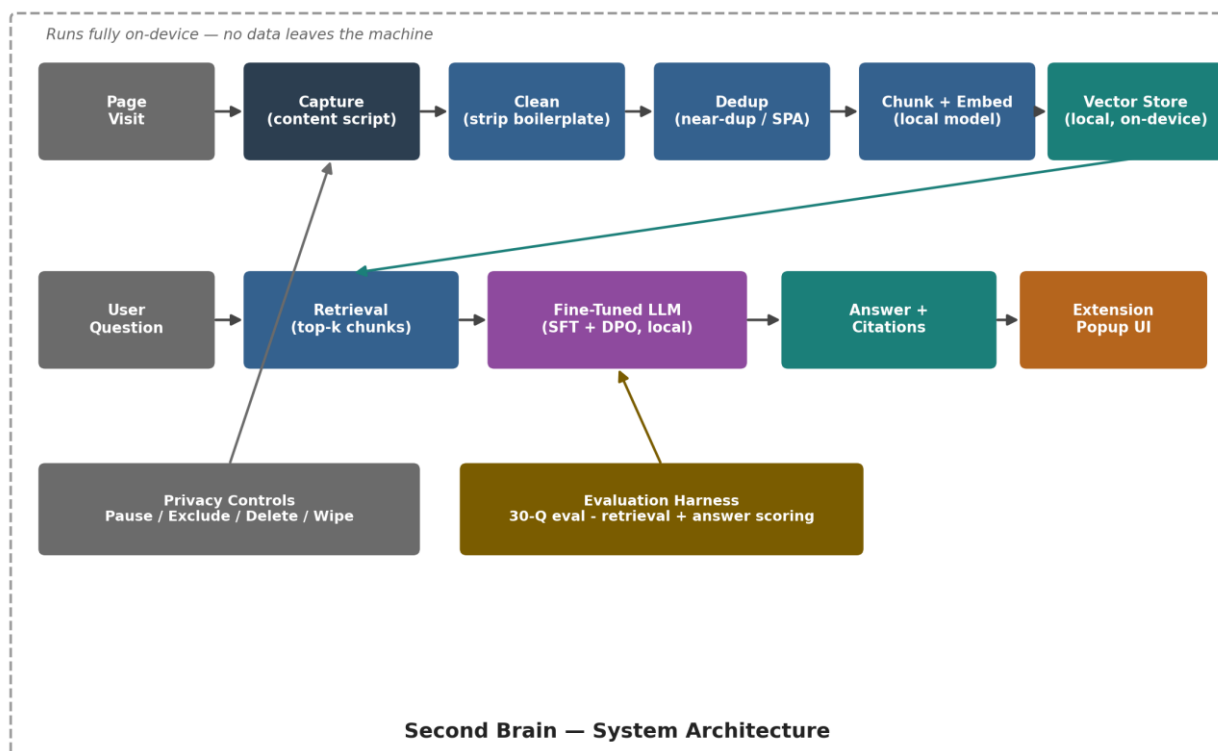
A Manifest V3 Chrome extension passively captures the readable content of visited pages under a strict, user-controlled allow/block model. Captured content is stripped of navigation, advertisements, and template boilerplate; near-duplicate visits and repeated single-page-application (SPA) states are collapsed; and the resulting clean text is chunked, embedded locally, and stored in an on-device vector index.

3.2 Fine-Tuned Generation Layer

A small open-weight language model is fine-tuned specifically for the narrow task of answering natural-language questions strictly from retrieved personal-history chunks, with inline citations back to the exact source and visit date. The model is trained in two stages: Supervised Fine-Tuning (SFT) to establish the grounded-answer-with-citation behavior, followed by Direct Preference Optimization (DPO) to explicitly reward correct “not found” refusals over confident but ungrounded answers — using the project’s own evaluation failures as training signal.

4. System Architecture

The diagram below summarizes the end-to-end pipeline. All components run locally; no stage transmits data off-device.



4.1 Pipeline Stage Breakdown

Stage	Responsibility	Key Design Concern
Capture	Content-script grabs readable DOM content on page visit, gated by allow/block list.	Minimal permission footprint; no capture on blocked domains.

Stage	Responsibility	Key Design Concern
Clean	Strips navigation, ads, and template boilerplate down to the article/content body.	Parsing fidelity varies heavily across site templates.
Dedup	Collapses re-visits, near-identical pages, and repeated SPA route states.	Hash/embedding-similarity based near-duplicate detection.
Chunk + Embed	Splits cleaned text into semantically coherent chunks; embeds locally.	Chunk size trade-off between retrieval precision and context completeness.
Vector Store	Persists embeddings and metadata (URL, title, timestamp) on-device.	IndexedDB-backed local store; no server dependency.
Retrieval	Given a query, returns top-k relevant chunks with source metadata.	Recall vs. precision tuning of k and similarity threshold.
Generation	Fine-tuned LLM answers strictly from retrieved chunks, with citations.	Grounding fidelity and calibrated refusal are the primary metrics.
Control Panel	Pause capture, exclude a site, delete an entry, or wipe the full index.	Every control must be reachable in one click.

5. Key Features

Feature	Description
Passive local capture	Indexes pages as they are naturally read, with a strict user-defined allow/block list.
Boilerplate-free extraction	Reduces pages to their real content before chunking, improving retrieval quality.
Near-duplicate & SPA dedup	Prevents re-visits and dynamic route changes from polluting the index.
Fully local embeddings & storage	No page content ever leaves the device at any pipeline stage.
Fine-tuned local generation model	SFT + DPO trained specifically for grounded answering and calibrated refusal.
Citation-grounded answers	Every answer links back to the exact source page and the date it was read.
Calibrated “not found” behavior	Explicitly trained to refuse rather than fabricate when context is insufficient.
Multi-hop question answering	Can combine information retrieved from two or more distinct pages.
Time-scoped queries	Supports questions such as “what did I read about X last week.”
One-click privacy controls	Pause, exclude a site, delete a single entry, or wipe the entire index instantly.
Zero cost, zero cloud	Runs fully offline on consumer hardware with no recurring infrastructure cost.
Quantized on-device inference	4-bit quantized model for low-latency, low-memory local generation.

6. Technical Stack

Layer	Technology
Extension	Manifest V3, TypeScript, Chrome Extension APIs (content scripts, background service worker)
Content extraction	Readability-style parser with per-domain boilerplate handling
Embedding model	Local sentence-embedding model (e.g. bge-small / nomic-embed), run on-device

Layer	Technology
Vector store	IndexedDB-backed local vector index
Generation model (base)	Small open-weight instruct model (e.g. Qwen2.5-3B/7B-Instruct or Llama-3.2-3B)
Fine-tuning	PyTorch + Hugging Face PEFT (LoRA / QLoRA) for SFT; TRL library for DPO
Quantization & local serving	GGUF 4-bit quantization via llama.cpp / Ollama for on-device inference
Evaluation harness	Python scoring scripts against the curated question set, run per configuration

7. Fine-Tuning Strategy

This is the component that elevates the project from a working RAG pipeline to a genuine applied-research contribution: rather than relying on a general-purpose model at inference time, the generation model is deliberately specialized for this system's exact failure modes.

7.1 Objective

Train a small local language model to (a) answer strictly from retrieved context with an inline citation to the source, and (b) explicitly refuse with “not found in your history” when the retrieved context does not support an answer — rather than defaulting to confident, ungrounded generation.

7.2 Data Pipeline

- **Seed set:** the 30 hand-verified questions (with ground truth and source location) built from the real two-week browsing corpus.
- **Synthetic expansion:** additional paraphrases, multi-hop combinations, and negative (“never read”) cases generated from chunks actually retrieved from the real corpus, then manually spot-checked for grounding accuracy before use.
- **Preference pairs for DPO:** mined directly from the base model's own evaluation run — each case where the base model produced a wrong or hallucinated answer becomes a “rejected” sample, paired with the verified correct, cited answer as “chosen.”

7.3 Training Recipe

- **Stage 1 — SFT:** LoRA/QLoRA fine-tuning on (question + retrieved chunks) → grounded answer with citation, or explicit refusal. Trains the base task shape before any preference optimization.
- **Stage 2 — DPO:** Preference optimization on chosen/rejected pairs mined from real evaluation failures, directly rewarding correct refusals over confident hallucination and correct citations over uncited claims.
- **Stage 3 — Quantization:** The fine-tuned model is quantized to 4-bit (GGUF) for fast, low-memory local inference through llama.cpp, with latency and memory measured before and after quantization.

7.4 Expected Outcome

At this model scale and data volume, the realistic goal is a measurable reduction in hallucination rate and materially better-calibrated refusals on the negative test cases — not a categorical accuracy jump. This is reported honestly in the evaluation results rather than overstated, since precise, defensible claims read far better in review than inflated ones.

8. Evaluation Methodology

Evaluation is treated as a first-class deliverable, not an afterthought, in line with the project's original design brief.

8.1 Evaluation Set

A hand-built set of 30 questions drawn from the real two-week browsing corpus, each with a verified ground-truth answer and exact source location, spanning four categories:

- Single-hop — answerable from one retrieved page.
- Multi-hop — requires combining information across two or more pages.
- Negative — concerns topics never actually read; correct behavior is explicit refusal.
- Time-scoped — requires reasoning over when something was read (e.g. “last week”).

8.2 Scoring Rubric

Each answer is scored as Correct, Partial, Wrong, or Hallucinated, tracked separately for retrieval quality and generation quality so that failures can be attributed to the right pipeline stage.

8.3 Comparative Evaluation

Retrieval is held fixed while the generation model is swapped between the base (untuned) model and the fine-tuned model, isolating the effect of fine-tuning. Metrics tracked: hallucination rate, refusal precision/recall on negative cases, multi-hop accuracy, end-to-end latency, and memory footprint before and after quantization.

8.4 Error Analysis

The five most interesting documented failures — with their retrieved chunks — are analyzed explicitly, covering causes such as boilerplate leakage into chunks, missed near-duplicate collapsing, half-loaded SPA capture, and genuine retrieval-versus-generation misattribution.

9. Privacy & Threat Model

Because this system indexes a person's real reading history, privacy is treated as a core design constraint rather than an add-on.

- No page content, embeddings, or model weights are transmitted off-device at any stage.
- Capture is opt-out by domain by default sensitivity categories (e.g. banking, private webmail) and fully user-configurable via an allow/block list.
- One-click controls: pause capture, exclude a specific site, delete a single indexed entry, or wipe the entire index.
- Threats considered: extension permission creep, exposure of the local index if the device itself is compromised, and accidental capture of sensitive form-based pages — mitigated through conservative default blocklists and content-type heuristics.
- Any published evaluation data (questions, ground truths, logs) is manually reviewed and redacted of personally sensitive content before inclusion in the repository.

10. Development Roadmap (6 Months)

Month	Milestone
Month 1	Harden capture/clean/dedup pipeline; continue real corpus collection; re-verify the existing 30-question baseline eval on the base (untuned) model.
Month 2	Literature review on grounded QA and RAG hallucination mitigation; build the synthetic data expansion pipeline; select and set up the base model and local training environment.
Month 3	Run SFT training; iterate on data quality and hyperparameters based on evaluation feedback.
Month 4	Mine preference pairs from evaluation failures; run DPO training; comparative evaluation of base vs. fine-tuned model.
Month 5	Quantize the fine-tuned model; optimize on-device inference; benchmark latency and memory; polish extension UI/UX.
Month 6	Final error analysis, FINDINGS documentation, demo preparation, written report, and buffer for edge cases.

11. Future Scope

- Multimodal extension: capture and index images, diagrams, and PDFs encountered while browsing, not just text — a natural bridge toward vision-language model experience.
- Personal knowledge graph layer connecting entities and topics across reading history over time, enabling relationship-aware queries.
- Proactive resurfacing: surfacing related material a user previously read when it becomes newly relevant.
- Periodic on-device re-fine-tuning as the reading corpus grows, keeping the model calibrated to the user's evolving content mix.
- End-to-end encrypted, local-first cross-device sync, preserving the zero-cloud-plaintext guarantee.
- Support for additional personal sources beyond the browser: PDFs, EPUBs, and podcasts (via local ASR) feeding the same index.

12. Conclusion

Second Brain demonstrates that a genuinely useful, trustworthy personal memory system does not require cloud infrastructure or a general-purpose foundation model used as-is. By combining a carefully engineered local capture-and-retrieval pipeline with a generation model deliberately fine-tuned — via SFT and DPO — for grounded answering and calibrated refusal, and by validating the entire system against a real, uncured personal corpus rather than a convenient public benchmark, this project delivers both a working product and a defensible applied-research contribution: a documented, measured improvement in exactly the failure mode that matters most for a personal memory system — knowing the difference between what it actually knows and what it doesn't.